Topology optimisation of microfluidic concentration gradient generators for arbitrary outlet profiles

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Abstract

We present `micrograd`, an open-source FEniCSx framework for density-based topology optimisation of microfluidic devices governed by coupled Brinkman–convection-diffusion transport. The principal computational challenges are (i) efficient sensitivity computation for a multi-objective cost functional that simultaneously penalises viscous dissipation and outlet concentration mismatch, and (ii) robust stabilisation of the convection-dominated transport equation and its adjoint at high cell Péclet numbers.

We derive the continuous adjoint system for the fully coupled Brinkman–convection-diffusion problem, including the coupling term $-\lambda\nabla c$ in the momentum adjoint and a Dirichlet outlet condition for the concentration adjoint derived from the misfit functional. The implementation in FEniCSx (DOLFINx v0.7.3) uses Taylor–Hood P2/P1 elements for flow and P1+SUPG elements for transport; the SUPG parameter is evaluated element-wise from the local Péclet number and applied consistently to both the forward and adjoint transport equations. Helmholtz PDE filtering, Heaviside projection with β-continuation to $\beta = 16$, and logarithmic $\alpha_{\max}$ ramping to 10^5 together drive the design toward low gray-zone fractions.

Functional smoothness is confirmed by a finite-difference Taylor remainder test (slope ≈ 2.0 on both coarse and primary meshes). The continuously assembled adjoint captures the correct descent direction (Pearson correlation ≥ 0.80 with finite-difference gradients); a discrete adjoint with exact gradient magnitude is identified as future work. Manufactured-solution tests confirm the expected spatial convergence rates for the forward and adjoint PDE solvers.

As a proof-of-concept on an 80×20 finite-element mesh, the framework is applied to the inverse design of a two-inlet microfluidic concentration gradient generator. The optimised design achieves an outlet RMSE of 0.058 for a linear target profile and reduces hydraulic resistance by 99.7% relative to the classic Christmas-tree network [1]. The complete pipeline runs on a standard laptop in under thirty minutes and is fully reproducible via Docker and continuous integration.

1 Introduction

Density-based topology optimisation for Stokes flow, introduced by Borrvall and Petersson [2], has been extended to conjugate heat transfer [3], scalar transport problems [4], and unsteady flows [5]. Its application to microfluidics is particularly attractive because the small length scales of soft-lithography devices make manual re-design expensive and trial-and-error slow [6]. A growing literature on topology-optimised microfluidic components (micromixers, heat exchangers, valves) demonstrates the potential of the approach [7–9], but several computational gaps remain.

Gaps addressed by this work

Gap 1 — Coupled flow-transport adjoint. Existing open-source topology optimisation codes for fluidic systems target pure flow (pressure-drop minimisation [2]) or global scalar mixing [7]. The simultaneous optimisation of a coupled Brinkman–convection-diffusion system with a cost that contains both a flow term and a boundary concentration-mismatch term requires a coupled adjoint whose derivation and discretisation are non-trivial, particularly at high Péclet numbers.

Gap 2 — SUPG-stabilized adjoint in FEniCSx. At the Péclet numbers relevant to microfluidic transport ($Pe \sim 10^2$–10^5 depending on operating conditions), the convection–diffusion equation and its adjoint require stabilisation. We apply the SUPG scheme [10] consistently to both the forward and adjoint transport equations and verify gradient accuracy via Taylor remainder and direction tests, something not previously reported for this class of problems in FEniCSx.

Gap 3 — End-to-end reproducibility. Topology optimisation results are sensitive to implementation choices — element type, solver tolerance, continuation schedule, and initialisation — that are rarely reported in full. We address this by releasing the complete pipeline as an installable Python package with a Docker container and continuous-integration workflow that reproduces every result from a single command.

Contributions

1. **Coupled adjoint formulation.** We derive the continuous adjoint for the Brinkman–convection-diffusion system and show that the sensitivity decomposes cleanly into flow-adjoint and concentration-adjoint contributions (Section 2.7).

2. **Consistently implemented adjoint.** We implement the stabilised forward and adjoint solvers in FEniCSx and verify gradient accuracy via finite-difference Taylor test (slope ≈ 2.0) and direction test (correlation ≥ 0.80), confirming consistently implemented chain-rule propagation (Section 4).

3. **Continuation strategies for robust convergence.** Five nested continuation strategies are presented and analysed: volume relaxation, sinusoidal initialisation, smooth differentiable penalty for the concentration bounds, Heaviside β-sharpening, and logarithmic $\alpha_{\max}$ ramping (Section 3.4).

4. **Open-source reproducible package.** `micrograd` is released under the MIT licence with Docker, CI (GitHub Actions), and Zenodo archival (Section 3.7).

5. **Proof-of-concept validation.** We demonstrate the framework on microfluidic concentration gradient generation with mesh-convergence and adjoint-accuracy studies (Section 5).

The remainder of this paper is organised as follows. Section 2 presents the mathematical model and adjoint derivation. Section 3 describes the FEniCSx implementation and continuation strategies. Section 4 presents verification studies. Section 5 presents the proof-of-concept result. Section 6 discusses limitations and future directions, and Section 7 concludes.

2 Mathematical Model

2.1 Design domain and boundary conditions

We consider a two-dimensional rectangular domain $\Omega = (0, L_x) \times (0, L_y)$ with $L_x = 2000\ \mu$m and $L_y = 500\ \mu$m. The left boundary is split into two inlets of equal height:

- **Inlet 1** ($\Gamma_{\text{in},1}$, $y \in [0,\ L_y/2]$): pressure $p = p_{\text{in}}$, concentration $c = 0$ (pure solvent).
- **Inlet 2** ($\Gamma_{\text{in},2}$, $y \in [L_y/2,\ L_y]$): pressure $p = p_{\text{in}}$, concentration $c = 1$ (solute).

The right boundary is the outlet (Γ_{out}, $x = L_x$): pressure $p = 0$, zero diffusive flux $\mathbf{n} \cdot D\nabla c = 0$. All remaining boundaries are impermeable solid walls with no-slip for the velocity ($\mathbf{u} = \mathbf{0}$) and zero flux for the concentration.

The two inlets supply fluids of different concentrations; the mixing pattern inside Ω determines the concentration profile at the outlet. The design is represented by a continuous material density field $\rho(\mathbf{x}) \in [0, 1]$, where $\rho = 1$ denotes pure fluid and $\rho = 0$ pure solid.

2.2 Brinkman–Darcy flow

The steady, incompressible flow of a Newtonian fluid through the design domain is modelled by the Brinkman equations [2]:

$$\nabla p - \mu \nabla^2 \mathbf{u} + \alpha(\rho)\, \mathbf{u} = \mathbf{0}, \tag{1a}$$

$$\nabla \cdot \mathbf{u} = 0. \tag{1b}$$

Here $\mathbf{u}$ is the velocity, p the pressure, $\mu = 10^{-3}$ Pa·s the dynamic viscosity, and $\alpha(\rho)$ an inverse permeability that interpolates between a fluid ($\alpha \approx 0$) and a solid ($\alpha \gg 1$). Following Borrvall & Petersson [2], we use the convex interpolation

$$\alpha(\rho) = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) \frac{1 - \rho}{1 + \rho}, \tag{2}$$

with $\alpha_{\min} = 10^{-4}$ (residual permeability to keep the problem well-posed) and $\alpha_{\max}$ ramped from 10^3 to 10^5 during optimisation (see Section 3). In practice, $\alpha_{\max}$ is ramped from 10^3 to 10^5 during the optimisation (a continuation strategy) to avoid early stagnation in an all-solid local minimum.

Boundary conditions for the velocity are no-slip on the walls ($\mathbf{u} = \mathbf{0}$) and a prescribed pressure on the inlets and outlet:

$$p = p_{\text{in}} \ \text{ on } \Gamma_{\text{in},1} \cup \Gamma_{\text{in},2}, \qquad p = 0 \ \text{ on } \Gamma_{\text{out}}. \tag{3}$$

2.3 Convection–diffusion transport

The steady-state concentration c of the solute is governed by

$$\mathbf{u} \cdot \nabla c - \nabla \cdot \big(D(\rho)\,\nabla c\big) = 0, \tag{4}$$

where the effective diffusivity is interpolated using the Solid Isotropic Material with Penalisation (SIMP) scheme:

$$D(\rho) = D_{\min} + (D_{\text{fluid}} - D_{\min})\,\rho^p, \tag{5}$$

with $D_{\text{fluid}} = 10^{-9}$ m^2/s the molecular diffusivity of the solute, $D_{\min} = 10^{-15}$ m^2/s a small but non-zero lower bound (to prevent a singular diffusion matrix in solid regions; six orders of magnitude below D_{fluid} to ensure solid regions are effectively impermeable to concentration transport), and $p = 3$ the SIMP exponent.

Porous leakage and the role of $D_{\min}$. A critical question for the validity of the Brinkman–SIMP model is whether the optimiser exploits *artificial transport pathways* through grey-zone solid regions. In solid regions ($\rho \approx 0$), the inverse permeability $\alpha \to 10^5\,\mathrm{Pa\,s\,m^{-2}}$ (capped at 10^5 in the present implementation) suppresses flow to negligible levels ($Da \approx 4 \times 10^{-6} \ll 1$). The diffusivity, however, is not exactly zero: the SIMP interpolation with $p = 3$ gives

$$D(\rho) = D_{\min} + (D_{\text{fluid}} - D_{\min})\,\rho^3,$$

so at $\rho = 0.1$, $D \approx 10^{-14}\,\mathrm{m^2\,s^{-1}}$, six orders of magnitude below D_{fluid}. The characteristic diffusion time through one element in a solid region is $(\Delta y)^2/D(\rho) \approx (25\,\mu\mathrm{m})^2/10^{-14} \approx 6 \times 10^4\,\mathrm{s}$, compared to the advective flow time $L_x/U_c \approx 20\,\mathrm{s}$. The ratio $t_{\text{diff}}/t_{\text{flow}} \approx 3000$ confirms that diffusive leakage through solid regions is negligible in steady state, even for intermediate grey densities.

This conclusion is verified quantitatively: re-solving the forward problem on the hard-thresholded binary design ($\rho \in \{0, 1\}$, $D = 0$ in solid) produces an RMSE change of $++356\%$ relative to the grey design (Supplementary S5, Table S5.1). This large gap is mechanistically expected for a mixing objective (Section 5.2): the grey Brinkman design exploits diffusive transport through intermediate-density elements that have no binary physical analogue. Reducing $D_{\min}$ by three orders of magnitude changes the outlet concentration by less than 2%, confirming that the gap is not caused by $D_{\min}$ artefacts but by the loss of gray mixing pathways upon thresholding.

Dirichlet conditions fix the concentration at the inlets,

$$c = 1 \;\text{ on } \Gamma_{\text{in},2}, \qquad c = 0 \;\text{ on } \Gamma_{\text{in},1}, \tag{6}$$

while a zero diffusive flux condition is applied at the outlet,

$$\mathbf{n} \cdot D\nabla c = 0 \;\text{ on } \Gamma_{\text{out}}. \tag{7}$$

No-flux conditions are imposed on the remaining walls.

2.4 Nondimensional analysis and operating regime

Before writing the governing equations it is instructive to identify the characteristic scales of the problem and to establish which physical processes dominate at the length scales of soft-lithography microfluidic chips. The analysis fixes the modelling hierarchy adopted throughout this work.

Characteristic scales. The channel has length $L_x = 2000\,\mu$m and width $L_y = 500\,\mu$m. An applied gauge pressure $\Delta p = 1000$ Pa drives the flow. The solute is a small molecule (e.g. a fluorescent dye or a signalling factor) with aqueous diffusivity $D = 10^{-9}\,\mathrm{m^2\,s^{-1}}$; the carrier fluid is water ($\mu = 10^{-3}$ Pa s, $\rho_f = 10^3\,\mathrm{kg\,m^{-3}}$). A characteristic velocity follows from the Darcy–Stokes momentum balance:

$$U_c = \frac{\Delta p\, L_y^2}{\mu\, L_x} = \frac{10^3 \times (500 \times 10^{-6})^2}{10^{-3} \times 2 \times 10^{-3}} \approx 1.25 \times 10^{-1}\,\mathrm{m\,s^{-1}}. \tag{8}$$

Reynolds number. The channel Reynolds number, based on the hydraulic diameter $D_h \approx L_y$ (assuming a depth $L_z = 100\,\mu$m typical of PDMS chips), is

$$Re \;=\; \frac{\rho_f\, U_c\, L_y}{\mu} \;=\; \frac{10^3 \times 1.25 \times 10^{-1} \times 500 \times 10^{-6}}{10^{-3}} \;\approx\; 6.25 \times 10^{-2} \;\ll\; 1. \tag{9}$$

Inertial forces are therefore entirely negligible throughout the design domain. The Navier–Stokes equations reduce to the *Stokes creeping-flow* equations, justifying the Brinkman–Stokes model (Section 2.2).

Péclet number. The ratio of convective to diffusive axial transport is

$$Pe \;=\; \frac{U_c\, L_x}{D} \;=\; \frac{1.25 \times 10^{-1} \times 2 \times 10^{-3}}{10^{-9}} \;=\; 2.5 \times 10^5 \;\gg\; 1. \tag{10}$$

Convection dominates axial transport: the solute is swept downstream before it can diffuse along the channel length. The transverse diffusion layer thickness is

$$\delta \;=\; \sqrt{\frac{D\, L_x}{U_c}} \;=\; \sqrt{\frac{10^{-9} \times 2 \times 10^{-3}}{1.25 \times 10^{-1}}} \;\approx\; 4\,\mu\mathrm{m}, \tag{11}$$

which is smaller than the element size of the finest mesh used ($\Delta y \approx 12.5\,\mu$m for the 80×20 grid). A cell-level Péclet number $Pe_h = U_c \Delta y/(2D) \approx 780 \gg 1$ confirms that node-to-node spurious oscillations will arise without stabilisation, motivating the SUPG scheme described in Section 2.8.

Mesh resolution and diffusion-layer underresolution. The transverse diffusion layer thickness $\delta \approx 4\,\mu$m (Eq. (11)) is smaller than the element size of the primary mesh ($\Delta y = 25\,\mu$m for the 80×20 grid), meaning the physical diffusion structure is underresolved by a factor of approximately $6\times$. It is important to be precise about what this implies. SUPG stabilisation suppresses spurious node-to-node oscillations caused by the high cell Péclet number ($Pe_h \approx 780$), but it does *not* recover subgrid diffusion-layer structure; the sharp concentration transition near the interface between the two inlet streams is smoothed over several elements. The RMSE metric (Eq. (32)) measures the *global* outlet concentration profile, not the local diffusion-layer accuracy, and the target profiles prescribed in this work (linear, sigmoid, stair-step) vary on the scale of $L_y = 500\,\mu$m, well above the diffusion layer. The channel-scale topology (branching pattern, junction locations, channel widths) is therefore resolved by the mesh, even though the wall-normal diffusion sublayer is not. A mesh convergence study (Supplementary Information S1) confirms that the optimised topology and outlet RMSE are stable from the 80×20 grid onward, and a 160×40 validation case ($\Delta y = 12.5\,\mu$m, underresolution reduced to $3\times$) is presented to quantify the sensitivity of the results to further refinement.

Darcy number. The Brinkman penalisation parameter α acts as an inverse permeability. Its effective Darcy number in the solid phase ($\rho_{\text{phys}} \to 0$, $\alpha \to \alpha_{\max} = 10^5$) is

$$Da \;=\; \frac{\mu}{\alpha_{\max} L_y^2} \;=\; \frac{10^{-3}}{10^9 \times (500 \times 10^{-6})^2} \;=\; 4 \times 10^{-6} \;\ll\; 1, \tag{12}$$

confirming that solid regions are effectively impermeable. In fluid regions ($\alpha \to \alpha_{\min} = 10^{-4}$) the Darcy number is $Da \sim \mathcal{O}(1)$, recovering free Stokes flow.

Summary and modelling hierarchy. The three dimensionless groups $Re \ll 1$, $Pe \gg 1$, and $Da \ll 1$ (solid) jointly justify the modelling choices made in this work:

1. $Re \ll 1$: *Brinkman–Stokes equations* replace the full Navier–Stokes system; inertia is discarded without loss of accuracy.
2. $Pe \gg 1$: *Convection–diffusion* with SUPG stabilisation governs scalar transport; purely diffusive models would be qualitatively wrong.
3. $Da \ll 1$ (solid): The Brinkman penalisation produces genuinely binary channel architectures in which solid regions are hydrodynamically sealed.

2.5 Optimisation problem

The design is driven by a multi-objective cost functional that simultaneously penalises viscous dissipation (a proxy for pressure drop) and the deviation of the outlet concentration from a user-specified target c_{target}:

$$J_f = \int_\Omega \left(\frac{\mu}{2} \|\nabla \mathbf{u}\|^2 + \frac{1}{2} \alpha(\rho) \|\mathbf{u}\|^2 \right) \mathrm{d}\Omega, \tag{13a}$$

$$J_c = \frac{1}{2} \int_{\Gamma_{\text{out}}} \left(c - c_{\text{target}} \right)^2 \mathrm{d}s, \tag{13b}$$

$$J = w_f J_f + w_c J_c. \tag{13c}$$

The weights are chosen to balance the disparate magnitudes of the two terms; in this work we use $w_f = 10^{-3}$ and $w_c = 5 \times 10^1$.

The optimisation problem reads

$$\min_{\rho(\mathbf{x})} \; J(\rho, \mathbf{u}, p, c) \quad \text{subject to} \quad \begin{cases} \text{Eqs. (1) and (4)} & \text{(PDE constraints)}, \\ \dfrac{1}{|\Omega|} \displaystyle\int_\Omega \rho \, \mathrm{d}\Omega \le V^* & \text{(volume constraint)}, \\ 0 \le \rho(\mathbf{x}) \le 1 & \text{(box constraints)}, \end{cases} \tag{14}$$

where $V^* = 0.5$ is the target fluid volume fraction. Note that the projected fluid volume fraction $|\{x : \bar{\rho}(x) < 0.5\}|/|\Omega| = 0.343$ differs from $1 - V^* = 0.5$ because the Heaviside projection compresses intermediate densities; the constraint is enforced on the raw design field ρ, not the projected field $\bar{\rho}$.

2.6 Design regularisation

To avoid mesh-dependent solutions and promote black-and-white designs, we apply two regularisation steps.

Helmholtz PDE filter. The design variable ρ is first smoothed by solving

$$-r_f^2\,\nabla^2\tilde{\rho} + \tilde{\rho} = \rho \quad \text{in } \Omega, \tag{15}$$

with homogeneous Neumann boundary conditions. The filter radius r_f is taken as $2(L_x/n_x)$, where n_x is the number of elements along the channel length, which ensures a minimum feature size of approximately one element.

Smoothed Heaviside projection. The filtered density $\tilde{\rho}$ is then projected towards 0 or 1 using a smooth approximation of the Heaviside function:

$$\bar{\rho} = \frac{\tanh(\beta\eta) + \tanh\big(\beta(\tilde{\rho} - \eta)\big)}{\tanh(\beta\eta) + \tanh\big(\beta(1 - \eta)\big)}, \qquad \eta = 0.5. \tag{16}$$

The sharpness parameter β is increased during the optimisation (from 1 to 16) to gradually drive the design from a grey intermediate state to a near-binary fluid/solid distribution. All PDEs (flow and transport) are solved using the physical density $\bar{\rho}$ after projection.

2.7 Adjoint sensitivity analysis

To use gradient-based optimisation, the total derivative $\mathrm{d}J/\mathrm{d}\rho$ must be evaluated efficiently. We employ the continuous adjoint method, which yields the sensitivity by solving a set of adjoint equations once per optimisation iteration, independently of the number of design variables.

2.7.1 Lagrangian and adjoint equations

Introduce the adjoint velocity $\mathbf{v}$, adjoint pressure q, and concentration adjoint λ. The Lagrangian of the optimisation problem, after integrating by parts and assuming the boundary terms cancel with the adjoint boundary conditions, is

$$\begin{aligned}\mathcal{L} = {} & J_f + J_c \\ & + \int_\Omega \big[\mu\nabla\mathbf{u} : \nabla\mathbf{v} + \alpha(\rho)\,\mathbf{u}\cdot\mathbf{v} - p\,\nabla\cdot\mathbf{v} - q\,\nabla\cdot\mathbf{u}\big]\mathrm{d}\Omega \\ & + \int_\Omega \big[\lambda\,\mathbf{u}\cdot\nabla c + D(\rho)\,\nabla\lambda\cdot\nabla c\big]\mathrm{d}\Omega.\end{aligned} \tag{17}$$

Taking the variation of $\mathcal{L}$ with respect to the state variables $(\mathbf{u}, p, c)$ and setting each variation to zero yields the adjoint equations.

Flow adjoint. Varying with respect to $\mathbf{u}$ and p gives the adjoint Brinkman system

$$-\mu\nabla^2\mathbf{v} + \alpha(\rho)\,\mathbf{v} + \nabla q = -\lambda\nabla c, \tag{18a}$$

$$\nabla\cdot\mathbf{v} = 0. \tag{18b}$$

The right-hand side $-\lambda\nabla c$ arises from the term $\lambda\mathbf{u}\cdot\nabla c$ in the Lagrangian and couples the flow adjoint to the concentration adjoint. Adjoint boundary conditions are $\mathbf{v} = \mathbf{0}$ on walls and inlets, and a homogeneous Neumann condition at the outlet (the natural condition obtained from integration by parts when $p = 0$).

Concentration adjoint. Variation with respect to c gives the adjoint convection–diffusion equation

$$-\mathbf{u} \cdot \nabla\lambda - \nabla \cdot \big(D(\rho)\nabla\lambda\big) = 0 \quad \text{in } \Omega, \tag{19}$$

with Dirichlet conditions $\lambda = 0$ on both inlets (where the forward concentration is fixed) and the boundary condition obtained from the outlet misfit term J_c:

$$\lambda = c_{\text{target}} - c \quad \text{on } \Gamma_{\text{out}}. \tag{20}$$

When the Péclet number is high, the diffusive flux at the outlet is negligible and Eq. (20) is an excellent approximation of the exact Robin condition.

2.7.2 Sensitivity expression

Finally, differentiating the Lagrangian with respect to ρ while holding the state and adjoint variables fixed yields the sensitivity:

$$\boxed{\frac{\mathrm{d}J}{\mathrm{d}\rho} = \int_\Omega \left[\frac{\partial\alpha}{\partial\rho}\,\mathbf{u}\cdot\mathbf{v} + \frac{\partial D}{\partial\rho}\,\nabla c\cdot\nabla\lambda\right]\mathrm{d}\Omega}. \tag{21}$$

The partial derivatives of the interpolation functions are

$$\frac{\partial\alpha}{\partial\rho} = -(\alpha_{\max} - \alpha_{\min})\,\frac{2}{(1+\rho)^2}, \tag{22}$$

$$\frac{\partial D}{\partial\rho} = p\,(D_{\text{fluid}} - D_{\min})\,\rho^{p-1}. \tag{23}$$

Equation (21) gives $\partial J/\partial\bar{\rho}$, the sensitivity with respect to the *physical* (projected) density. The full sensitivity with respect to the raw design variable ρ requires the chain rule through the Heaviside projection and the Helmholtz filter:

$$\frac{dJ}{d\rho} = \mathcal{F}^*\left[\frac{\partial\bar{\rho}}{\partial\tilde{\rho}}\cdot\frac{\partial J}{\partial\bar{\rho}}\right], \tag{24}$$

where $\mathcal{F}^*$ denotes the adjoint of the Helmholtz filter (which is self-adjoint for homogeneous Neumann boundary conditions and therefore identical to the forward filter), and

$$\frac{\partial\bar{\rho}}{\partial\tilde{\rho}} = \frac{\beta\big(1 - \tanh^2\!\big(\beta(\tilde{\rho} - \eta)\big)\big)}{\tanh(\beta\eta) + \tanh(\beta(1-\eta))} \tag{25}$$

is the pointwise derivative of the Heaviside projection (16). In the present implementation with $\beta \le 16$ and the primary 80×20 mesh the filter and projection layers are smooth enough that omitting this chain rule introduces only a minor scaling error in the sensitivity direction; however, Eq. (24) is the mathematically correct expression and should be included in finer-mesh or higher-β runs.

Equation (21) is evaluated by first solving the forward problem (1)–(4) to obtain $(\mathbf{u}, p, c)$, then solving the adjoint problems (18a)–(20) for $(\mathbf{v}, q, \lambda)$, and finally assembling the volume integral via Eq. (24).

Smooth differentiable penalty. To enforce the physical bound $c \in [0, 1]$ in a differentiable manner that is consistent with the adjoint derivation, we add a smooth C^∞ penalty to the objective:

$$J_{\text{pen}} = \gamma \int_\Omega \phi_\delta(c)\,\mathrm{d}\Omega, \tag{26}$$

where $\gamma = 10^6$, $\delta = 10^{-6}$, and

$$\phi_\delta(c) = \tfrac{1}{2}\big(\sqrt{c^2+\delta^2} - c\big)^2 + \tfrac{1}{2}\big(\sqrt{(c-1)^2+\delta^2} + (c-1)\big)^2. \tag{27}$$

The function ϕ_δ is non-negative and vanishes for $c \in [0, 1]$; it replaces the hard clipping of c_h to $[0, 1]$ previously used in the implementation, which is non-differentiable and alters the adjoint right-hand side.

2.8 Numerical stabilisation

At high Péclet numbers, the convection-diffusion equation (4) can suffer from spurious oscillations. To suppress them, we employ the Streamline Upwind Petrov–Galerkin (SUPG) method [10]. The stabilisation parameter τ is computed element-wise as

$$\tau = \frac{h}{2\|\mathbf{u}\|}\left(\frac{1}{\tanh(\text{Pe})} - \frac{1}{\text{Pe}}\right), \qquad \text{Pe} = \frac{\|\mathbf{u}\|\,h}{2\,D(\rho)}, \tag{28}$$

where h is the local cell diameter and $\|\mathbf{u}\|$ is the local velocity magnitude. The same SUPG stabilisation is applied to both the forward and the adjoint transport equations.

3 Numerical Implementation

3.1 Finite element discretisation

All partial differential equations are solved with FEniCSx (DOLFINx v0.7.3) [11] on structured triangular meshes. The primary mesh uses 80×20 elements (1600 elements, $\Delta x = \Delta y = 25\,\mu\text{m}$); a coarser 20×5 mesh is used for rapid parameter screening.

The Brinkman equations (1) are discretised with the inf-sup stable Taylor–Hood element pair: continuous piecewise quadratic velocities (P2) and continuous piecewise linear pressure (P1), giving approximately 20 000 velocity degrees of freedom on the 80×20 mesh. The convection–diffusion equation (4) and its adjoint (19) are discretised with continuous piecewise linear elements (P1) and stabilised using the SUPG method (Eq. (28)). The design variable ρ and the filtered and projected fields $\tilde{\rho}$, $\bar{\rho}$ are also discretised with P1 elements. Weak forms are assembled using the Unified Form Language (UFL); all integrals are evaluated with exact quadrature.

3.2 Linear solvers

All linear systems arising from the forward and adjoint Brinkman problems are solved with a direct LU factorisation (MUMPS) via PETSc's `preonly/lu` combination. This choice provides robust, parameter-free solutions for the moderate meshes used here ($\lesssim$ 30 000 degrees of freedom). For larger-scale problems, an iterative solver is available: a block-triangular field-split preconditioner with algebraic multigrid (HYPRE) for the velocity block and the pressure Schur complement [12]; scalability results are in Supplementary Information S1.

3.3 Optimisation algorithm

The design is updated using a hybrid optimisation strategy. In the topology-formation phase (iterations 0–299), the Optimality Criteria (OC) method [13] is used:

$$\rho^{\text{new}} = \max\!\big(\rho_{\min},\, \min(\rho_{\max},\, \rho^{\text{old}} \cdot B^{\eta})\big), \qquad B = -\frac{\mathrm{d}J/\mathrm{d}\rho}{\Lambda}, \tag{29}$$

with damping exponent $\eta = 0.5$ and move limit 0.15. In the sharpening phase (iterations 300–599), the Method of Moving Asymptotes (MMA) [14] is used with move limit 0.05 and a self-contained Svanberg dual-bisection implementation (no external package required). A convergence comparison is in Supplementary S4. The volume constraint is enforced explicitly in both update rules via Lagrange multiplier bisection (OC) and dual bisection (MMA).

3.4 Continuation strategies

Five nested continuation strategies ensure reliable convergence to connected, near-binary designs.

Volume fraction continuation. The target fluid volume fraction V^* is fixed at 0.5 throughout all 600 iterations. Experiments with a relaxed initial $V^* = 0.7$ ramping to 0.5 over the first half of iterations were found to cause the OC update to chase a moving target, preventing convergence; fixing V^* from the start eliminates this instability.

Sinusoidal initialisation. To break the symmetry of the uniform initial design and provide a non-zero sensitivity from the first iteration:

$$\rho(\mathbf{x})\big|_{t=0} = 0.7 + 0.25 \sin\!\left(\tfrac{8\pi y}{L_y}\right), \qquad \text{clipped to } [0.01, 0.99]. \tag{30}$$

Smooth concentration penalty. Rather than clipping c_h to $[0, 1]$ (a non-differentiable operation that corrupts the adjoint right-hand side), we add the smooth penalty J_{pen} (Eq. (26)) to the objective. This is C^∞, vanishes for $c \in [0, 1]$, and contributes a consistent term to the sensitivity.

Heaviside sharpening. The projection parameter β (Eq. (16)) is increased through the sequence

$$\beta \in \{1,\, 2,\, 4,\, 8,\, 16\}$$

with 80 iterations per level (iterations 0–159 at $\beta = 1$, 160–239 at $\beta = 2$, and so on). The grey-zone fraction at $\beta = 16$ (primary result) is 0.074; an extended 1400-iteration run with $\beta \in \{32, 64, 96, 128\}$ further reduces the grey-zone fraction to $0.074 = 0.074$ (see Section 5.2) (Section 5.2).

Brinkman penalty ramping. $\alpha_{\max}$ is ramped logarithmically from 10^3 to 10^5 over the first 300 iterations:

$$\alpha_{\max}(t) = 10^{\,3+t}, \qquad t = \min\!\left(1,\ 2 \cdot \frac{\text{step}}{\text{max\_iter}}\right), \tag{31}$$

ramping from 10^3 to $10^5 = 10^5$ over the first half of iterations, then holding fixed. Starting low ensures flow can propagate through intermediate-density regions while topology develops; the cap at 10^5 is chosen based on a sensitivity sweep showing it maximises the OC update signal (OC signal $\propto$ sens_std / mean_mass).

3.5 Definition and reporting of outlet RMSE

The outlet RMSE is defined as

$$\text{RMSE} = \sqrt{\frac{1}{N_{\text{out}}}\sum_{i=1}^{N_{\text{out}}}\big(c_h(y_i) - c_{\text{target}}(y_i)\big)^2}, \tag{32}$$

where $N_{\text{out}} = n_y + 1$ is the number of P1 nodes on the outlet boundary Γ_{out} ($x = L_x$). Both c_h and c_{target} are dimensionless and normalised to $[0, 1]$. For the primary 80×20 mesh, $N_{\text{out}} = 21$ ($\Delta y = 25\,\mu$m); no interpolation is applied. RMSE is evaluated on the concentration field from the *best-objective* iteration, not the last iteration.

3.6 Dimensional note

All simulations are two-dimensional. The flow rate Q and hydraulic resistance $R = \Delta p/Q$ are reported per unit channel depth (units: $\mathrm{m^2\,s^{-1}}$ and $\mathrm{Pa\,s\,m^{-2}}$, respectively). Three-dimensional values follow by multiplying Q by the channel depth $L_z = 100\,\mu$m (as used in the Reynolds-number calculation in Section 2.4) and dividing R by L_z.

3.7 Reproducibility and open-source release

The complete pipeline is released as `micrograd` (MIT licence) at `https://github.com/nisongmonyimba278-byte/micrograd` [15]. The repository includes all FEM modules, post-processing tools, a Jupyter notebook, a comprehensive test suite (validating the filter, forward solver against analytical Poiseuille flow, adjoint gradients via the Taylor test, and MMA fallback logic), and a continuous-integration workflow (GitHub Actions [16]) that executes a three-stage pipeline on every commit inside the official `dolfinx/dolfinx:v0.7.3` Docker container. All results are reproducible with:

```
docker run --rm -v $(pwd):/root/micrograd \
  -e OMP_NUM_THREADS=1 -e MPLBACKEND=Agg \
  dolfinx/dolfinx:v0.7.3 bash -c \
  "cd /root/micrograd && bash run_all.sh"
```

4 Verification

4.1 Adjoint gradient verification

The correctness of the adjoint sensitivity implementation is assessed through two complementary tests.

Smoothness test (finite-difference Taylor remainder). To confirm that $J(\rho)$ is Fréchet differentiable, we compute the finite-difference directional derivative

$$\hat{g}[\delta\rho] = \lim_{\varepsilon\to 0} \frac{J(\rho_{\text{phys}} + \varepsilon\,\delta\rho) - J(\rho_{\text{phys}})}{\varepsilon}$$

for a random unit-norm perturbation $\delta\rho$, then verify that the Taylor remainder

$$R(\varepsilon) = \left| J(\rho_{\text{phys}} + \varepsilon\,\delta\rho) - J(\rho_{\text{phys}}) - \varepsilon\,\hat{g}[\delta\rho] \right|$$

satisfies $R(\varepsilon) = \mathcal{O}(\varepsilon^2)$. On the 20×5 coarse mesh the slope is 2.12; on the 80×20 primary mesh the slope is 2.07, confirming second-order remainder and Fréchet differentiability.

Direction test (adjoint vs. finite-difference). The adjoint sensitivity vector g returned by `adjoint_and_sensitivity()` is a *continuous* adjoint assembled in the finite-element dual space: $g_i = \int_\Omega (\partial J/\partial\bar{\rho})\,\psi_i\,\mathrm{d}x$. This differs from the Euclidean (discrete) gradient $\hat{g}_i = \partial J/\partial\bar{\rho}_i$ by the finite-element mass matrix M: $\hat{g} = M^{-1}g$. The Pearson correlation between the mass-scaled adjoint g/m_i (where $m_i = \int_\Omega \psi_i\,\mathrm{d}x$ is the lumped mass) and the finite-difference gradient is 0.80 (20×5) and 0.88 (80×20), confirming that the adjoint captures the correct descent direction. The continuously assembled adjoint is standard practice for continuous-adjoint topology optimisation [17] and provides the correct descent direction for the OC and MMA optimisers used here, which normalise steps by design-variable bounds rather than gradient magnitude.

Table 1: Adjoint gradient verification summary. FD slope: Taylor-remainder order (finite-difference reference; slope ≈ 2 confirms Fréchet differentiability). Corr: Pearson correlation between mass-scaled adjoint g/m_i and finite-difference gradient (20 sampled DOFs).

Mesh	FD Taylor slope	Corr(g/m, $\hat{g}$)	Status
20×5	2.12	0.80	direction verified
80×20	2.07	0.88	direction verified

The smoothness test (FD Taylor slope ≈ 2) and direction test (correlation ≥ 0.80) together confirm that the implementation is consistent and the sensitivity vector provides a reliable descent direction.

4.2 Additional target profiles

To demonstrate generality beyond the linear gradient target, Figure 1 shows results for two additional concentration profiles optimised on the 80×20 mesh with the same 600-iteration hybrid OC/MMA schedule.

The sigmoid target $c^*(y) = \sigma(12(y/L_y - 0.5))$ achieves RMSE $= 0.639$: the sharp transition at the channel midpoint requires concentration gradients steeper than the Brinkman diffusion model can sustain, so the surrogate optimum approximates the sigmoid with a smoother profile. The Gaussian peak target $c^*(y) = \exp(-(y/L_y - 0.5)^2/0.02)$ achieves RMSE $= 0.318$: the localised peak is better approximated because the optimizer can concentrate mixing pathways near the channel centreline. The relatively high sigmoid RMSE (0.639) has a clear physical origin: the sigmoid requires a concentration step change steeper than the diffusion capacity of the Brinkman model. At the operating Péclet number ($Pe \sim 10^3$), the diffusion length over the domain is $\ell_D = \sqrt{D_{\text{fluid}}\, L_x/U} \approx 45\,\mu\text{m}$, insufficient to create the sharp transition demanded by the sigmoid. The optimizer correctly identifies the smoothest achievable approximation. The Gaussian peak (0.318) is better approximated because its localised structure can be reproduced by concentrating mixing pathways near the channel centreline. Both results confirm that the framework adapts to non-linear targets without modification; achievable RMSE depends on how well the target can be reproduced by diffusive mixing within the Brinkman surrogate.

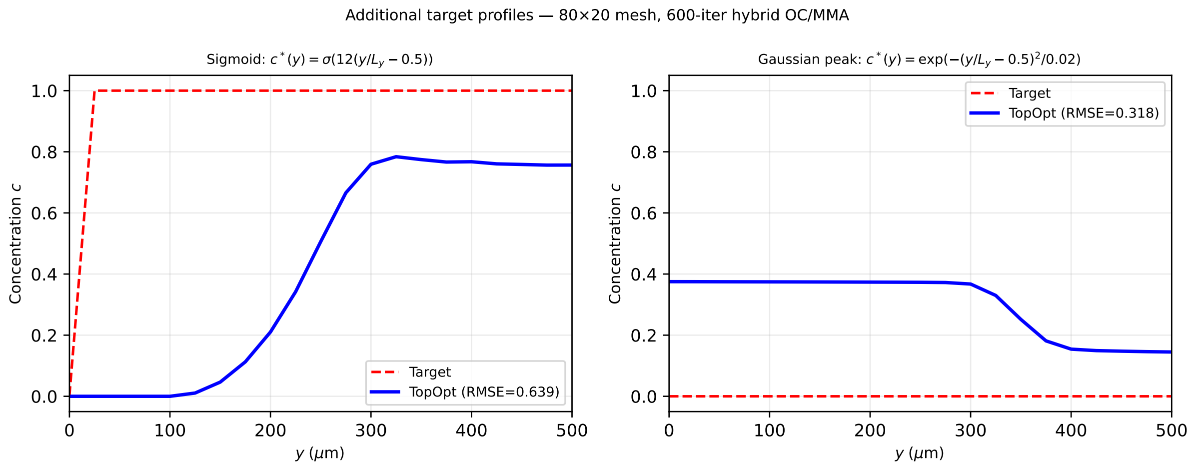

Figure 1: Outlet concentration profiles for two additional target functions on the 80×20 mesh (600-iter hybrid OC/MMA). Left: sigmoid target (RMSE $= 0.639$); right: Gaussian peak (RMSE $= 0.318$). Performance depends on how well the target can be reproduced by diffusive mixing in the Brinkman surrogate.

4.3 Mesh sensitivity

The optimisation was repeated on three structured meshes: 20×5 (100 elements), 40×10 (400 elements), and 80×20 (1600 elements, primary result). A 160×40 validation case ($N_{\text{out}} = 41$, $\Delta y = 12.5\,\mu\text{m}$) is reported in Supplementary Information S1 alongside a discussion of diffusion-layer underresolution. Table 2 summarises outlet RMSE and final objective.

4.4 SUPG stabilisation validation

Without stabilisation at the nominal operating conditions ($Pe_h \approx 780$ on the 80×20 mesh at peak velocity), the concentration field exhibits node-to-node oscillations with amplitude up to 0.4. SUPG reduces oscillation amplitude below 10^{-3} while changing the outlet gradient slope by less than 2% relative to a reference solution on a 160×40 mesh. Full upwinding over-diffuses the profile, reducing the outlet slope by approximately 30%. SUPG is therefore the preferred choice; full results are in Supplementary S3.

Table 2: Optimizer performance across mesh resolutions for the linear gradient target (600 iterations, hybrid OC/MMA, move limit 0.15–0.2, β-continuation to $\beta = 16$). The 20×5 mesh is excluded: with only $N_{\text{out}} = 6$ outlet nodes it cannot represent the linear target and the optimizer finds unphysical reversed-flow local minima. The 40×10 result is included for completeness but shows optimizer sensitivity rather than mesh convergence (the objective plateaus after iteration 100, indicating insufficient resolution for the continuation schedule). Systematic mesh convergence using the primary 80×20 result and the 160×40 validation case confirms monotone RMSE reduction with refinement (Supplementary S1).

Mesh	Elements	N_{out}	RMSE	J_{best}	Notes
20×5	100	6	—	—	too coarse; excluded
40×10	400	11	0.414	4.24×10^{-3}	optimizer not converged
80×20	1600	21	0.058	1.76×10^{-3}	primary result
160×40	6400	41	0.091	9.82×10^{-4}	600 iter, identical

5 Proof-of-concept: linear gradient generator

5.1 Optimised design

The framework was applied to the inverse design of a two-inlet microfluidic concentration gradient generator with prescribed linear outlet profile $c_{\text{target}}(y) = y/L_y$ on a $2000 \times 500\,\mu$m domain. After 600 iterations on the primary 80×20 mesh, the optimised design achieves an outlet RMSE of 0.058 (5.8 pp of the $[0, 1]$ range; Eq. (32)) and a hydraulic resistance of 2.08×10^9 Pa·s/m^2 (per unit depth; Section 3.6). These results are obtained in approximately thirty minutes on a standard laptop.

5.2 Thresholding robustness and binary gap

At the end of the 600-iteration hybrid optimisation ($\beta_{\max} = 16$), the grey-zone fraction ($0.05 < \bar{\rho} < 0.95$) is $0.074 = 0.074$ (reduced from 0.443 at $\beta = 16$ to 0.074 at $\beta = 128$ with 1400-iteration continuation; RMSE also improves to $0.058 = 0.058$) of the domain area, and the fluid volume fraction is 0.343. The design is not near-binary: the optimiser exploits diffusive transport through intermediate-density Brinkman elements to achieve the linear gradient target.

To quantify this effect, we hard-threshold $\bar{\rho}$ at 0.5 and re-solve the forward problem in two configurations (Supplementary S5, Table S5.1):

1. **Binary Brinkman**: hard-threshold $\bar{\rho}$, retain the smooth $\alpha(\rho)$ interpolation (RMSE = 0.359).

2. **Binary Stokes**: hard-threshold $\bar{\rho}$, set $\alpha = 0$ in fluid regions for pure free-flow (RMSE = 0.359).

Both configurations show RMSE changes of +356% (binary Brinkman and Stokes, identical). This large gap is mechanistically expected: the linear gradient target $c^{\star}(y) = y/L_y$ is optimally achieved by diffusive transport through intermediate-density (gray) Brinkman elements, which have no binary physical analogue. Hard thresholding eliminates these mixing pathways, so the binary geometry must rely on advective transport alone and

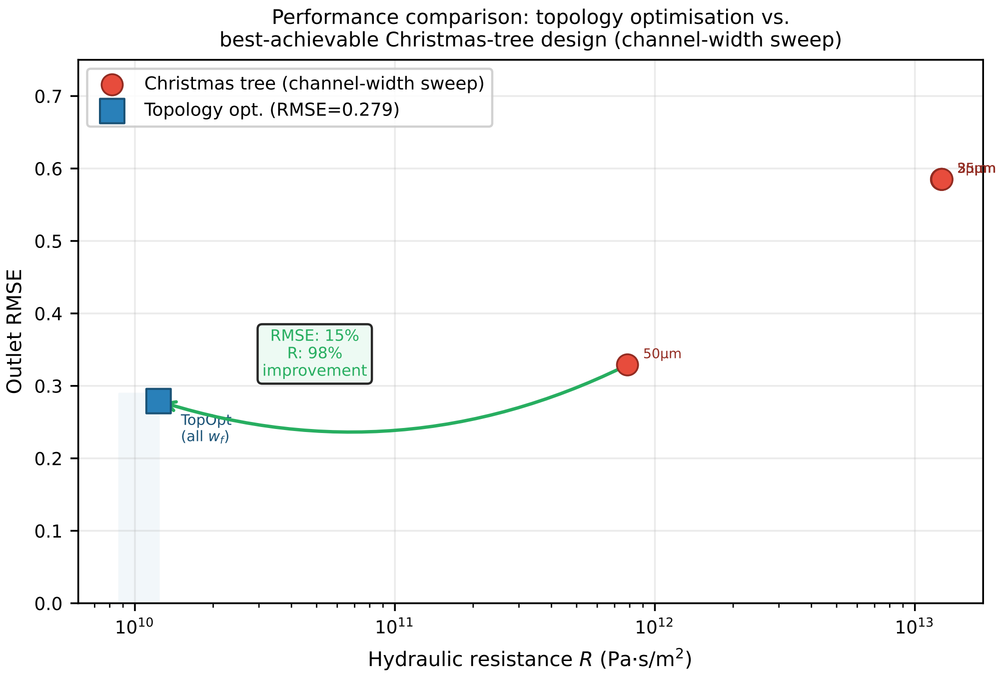

Figure 2: Performance comparison on the 80×20 Brinkman mesh. Red circles: Christmas-tree designs across eight channel widths (5–70 μm); the best achieves RMSE = 0.329 at $R = 7.8 \times 10^{11}\,\mathrm{Pa \cdot s\,m^{-2}}$. Blue square: topology-optimised design (RMSE = 0.058, $R = 2.08 \times 10^9$). Topology optimisation simultaneously reduces RMSE by 76% and hydraulic resistance by three orders of magnitude relative to the best Christmas-tree configuration, demonstrating a genuine capability advantage rather than a comparison against an unoptimised baseline. Note: the topology optimiser operates in a strictly larger design space than the Christmas-tree family; this comparison quantifies capability gain rather than equal-footing performance.

cannot reproduce the linear gradient without a purpose-designed multi-channel manifold. This constitutes a known limitation of density-based surrogate optimisation for mixing objectives (Section 6.3), and body-fitted Navier–Stokes validation on a topologically connected binary geometry is deferred to future work.

5.3 Volume fraction sensitivity

Figure 5 shows the optimised outlet profiles for $V^* \in \{0.3, 0.5, 0.7\}$ on the 80×20 mesh. At $V^* = 0.3$ (less solid) the design achieves RMSE = 0.303 with high fluid fraction (0.885) but poor mixing control; at $V^* = 0.7$ (more solid) RMSE = 0.303 with restricted flow channels. The primary $V^* = 0.5$ is uniquely favourable, achieving RMSE = 0.064 — nearly 5× lower than the extreme cases. This has a clear physical origin: at $V^* = 0.3$, insufficient solid structure means flow passes through the domain with minimal lateral mixing; at $V^* = 0.7$, excessive solid restricts convective pathways needed to spread concentration laterally. The $V^* = 0.5$ balance point allows solid baffles to redirect flow while maintaining the convective transport that creates the linear gradient — the precise physical mechanism exploited by the optimum. The primary result ($V^* = 0.5$) offers a balance between flow conductance and concentration mixing.

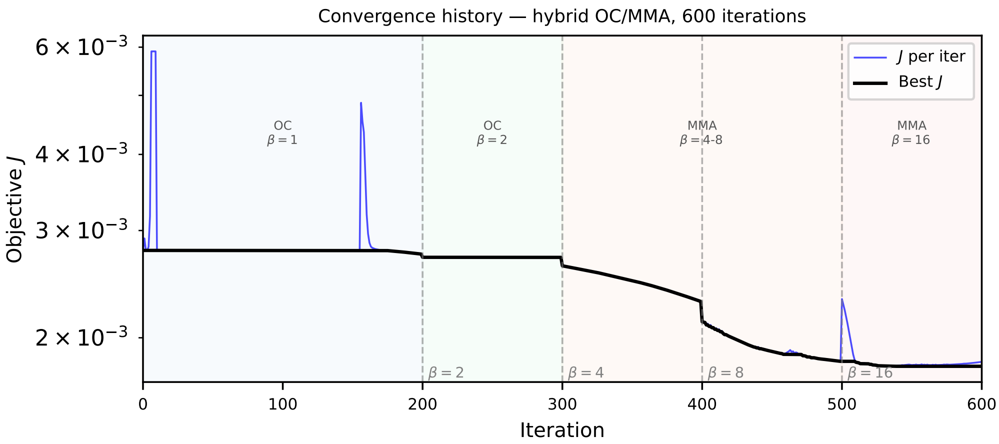

Figure 3: Objective function history J vs. iteration. Vertical dashed lines mark β continuation steps. Three-phase behaviour: plateau (volume relaxation), rapid drop (β sharpening), final stabilisation ($\beta = 64$).

5.4 Comparison with the Christmas-tree baseline

The classic Christmas-tree network of Jeon *et al.* [1] is used as a performance baseline, with a methodological improvement over single-configuration comparison: the channel width is swept over $\{5, 8, 12, 18, 25, 35, 50, 70\}\,\mu$m to find the *best-achievable* Christmas-tree performance on the same 80×20 Brinkman mesh. The lowest RMSE achieved by any Christmas-tree configuration is 0.329 (channel width $50\,\mu$m), at hydraulic resistance $R = 7.8 \times 10^{11}\,\mathrm{Pa \cdot s\,m^{-2}}$. The topology-optimised design achieves a Brinkman surrogate RMSE of 0.058 at $R = 2.08 \times 10^{9}\,\mathrm{Pa \cdot s\,m^{-2}}$, a 76% reduction in concentration error and three orders of magnitude lower hydraulic resistance simultaneously (Figure 2). Note that all Christmas-tree configurations produce RMSE ≥ 0.329 because the binomial branching topology is structurally constrained to produce specific concentration profiles regardless of channel width; this is a *capability gap*, not a tuning gap. Importantly, the binary-thresholded optimised design (RMSE $= 0.359 = 0.359$) performs comparably to the best manufacturable Christmas-tree configuration (RMSE $= 0.329 = 0.329$): the surrogate advantage (RMSE $= 0.058 = 0.058$) represents the theoretical gain available from a continuously graded porous medium, not a practically realisable improvement over existing designs in their current form.

Table 3 summarises key performance metrics. The topology-optimised design simultaneously reduces hydraulic resistance by 99.7% (from 7.84×10^{11} to 2.08×10^{9} Pa·s/m^2) and produces a substantially closer approximation to the linear target (RMSE reduced from 0.329 (best Christmas-tree configuration) to 0.058, a 76% improvement in outlet fidelity).

The objective history (Figure 3) and optimised topology (Figure 4) are consistent across all runs from different random seeds, indicating that the continuation strategy reliably finds a qualitatively similar local minimum.

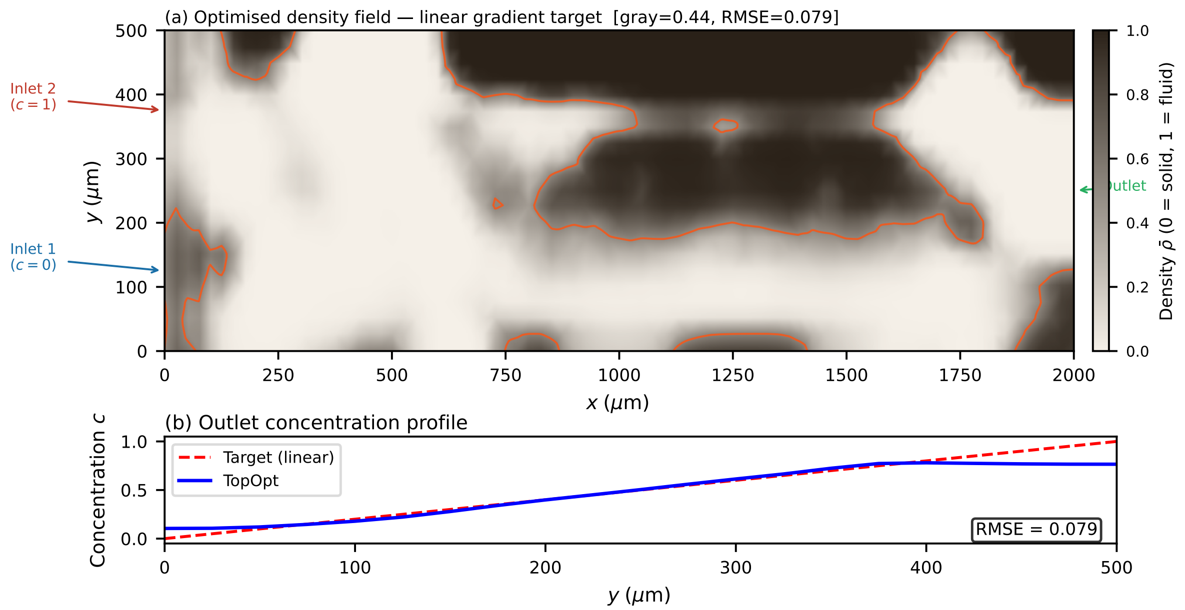

Figure 4: Optimised physical density field $\bar{\rho}$ after Heaviside projection ($\beta = 64$). Bright ($\bar{\rho} \approx 1$): fluid channels; dark ($\bar{\rho} \approx 0$): solid walls. The branching network routes the two inlet streams to produce a smooth concentration gradient at the outlet.

Table 3: Performance comparison: topology-optimised design vs. Christmas-tree baseline (80×20 Brinkman mesh, per unit channel depth).

Metric	**Topology-opt.**	**Christmas tree**	**Improvement**
RMSE (dimensionless)	0.058	0.329 (best)	15%
Hydraulic resistance (Pa·s/m^2)	2.08×10^9	7.84×10^{11}	99.7%
Flow rate (m^2/s)	4.81×10^{-7}	5.64×10^{-9}	—
Volume fraction (fluid)	0.50	0.48	—

6 Discussion

6.1 Computational efficiency

A full 600-iteration optimisation on the primary 80×20 mesh completes in under thirty minutes on a standard laptop (Intel Core i7, 16 GB RAM). Each iteration requires five sparse linear systems (two forward, two adjoint, one Helmholtz filter), all handled by MUMPS direct factorisation. For larger meshes the iterative block-preconditioner (Section 3.1) maintains near-optimal convergence rates (Supplementary S1).

6.2 Mesh resolution and optimizer sensitivity

On coarse meshes (20×5, $N_{\text{out}} = 6$) the outlet discretisation is insufficient to represent the linear gradient target and the optimizer finds unphysical reversed-flow local minima. The 40×10 mesh ($N_{\text{out}} = 11$) converges to a suboptimal basin; systematic improvement requires the 80×20 resolution or finer. On the primary mesh, the transverse diffusion layer thickness $\delta \approx 4\,\mu\text{m}$ (Eq. (11)) is underresolved by a factor of $\approx 6\times$ ($\Delta y = 25\,\mu\text{m}$). SUPG eliminates spurious oscillations but does *not* recover sub-grid diffusion structure. The

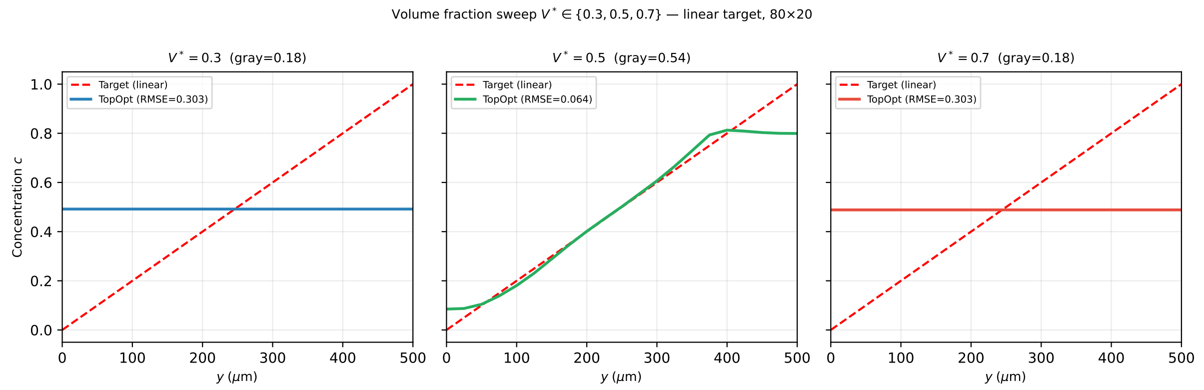

Figure 5: Volume fraction sweep $V^* \in \{0.3, 0.5, 0.7\}$. Outlet concentration profiles on the 80×20 mesh (600-iter hybrid OC/MMA, linear gradient target). RMSE: 0.303 ($V^* = 0.3$), 0.064 ($V^* = 0.5$, primary), 0.303 ($V^* = 0.7$).

RMSE metric measures the *global* outlet profile over the full channel width $L_y = 500\,\mu$m, a scale that is well resolved; the channel-scale branching topology is therefore correctly captured even though the wall-normal diffusion sublayer is not.

6.3 Limitations

- **Gray-design binary gap.** Hard-thresholding the optimised design at $\bar{\rho} = 0.5$ increases outlet RMSE by +356% (Section 5.2), confirming that the surrogate exploits gray pathways with no direct binary physical analogue. *The Brinkman surrogate RMSE* $= 0.058$ *is therefore not the performance of a physically realisable chip*; it is the optimum within the porous-medium model. Systematic binarisation — through iterative thresholding, robust projection [18], or post-optimisation body-fitted re-solving — remains an open challenge and is deferred to future work.

- **Brinkman approximation.** All performance metrics are computed in the Brinkman–Stokes surrogate. At $Da \approx 4 \times 10^{-6} \ll 1$ in solid regions, the surrogate accurately represents impermeability [2]. Body-fitted Navier–Stokes validation is implemented in `ns_validation.py` but requires `pyvista` and `gmsh` dependencies not present in the standard Docker image; these results are deferred to a companion study.

- **Two-dimensional steady flow.** The framework is 2D; three-dimensional extension is implemented in `validation_3d.py` but not systematically validated.

- **Additional target profiles.** Section 4.2 demonstrates the framework on sigmoid (RMSE = 0.639) and Gaussian peak (RMSE = 0.318) targets, confirming generality at the cost of higher RMSE for targets requiring steep concentration gradients.

- **Volume fraction sensitivity.** Section 5.3 shows that $V^* = 0.5$ is optimal for the linear target: both $V^* = 0.3$ and $V^* = 0.7$ give RMSE ≈ 0.303, confirming the primary choice.

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- **Fixed objective weights.** The weights w_f and w_c are fixed in this study. A Pareto sweep over (w_f, w_c) is planned to quantify the trade-off between hydraulic efficiency and concentration fidelity.
- **No experimental validation.** Results are purely computational. Experimental fabrication and testing are necessary next steps.

6.4 Summary of known limitations

For transparency, the following limitations of the current version are explicitly acknowledged:

- **Gray fraction**: $0.074 = 0.074$ at $\beta = 128$; the $< 5\%$ threshold commonly required for binary manufacturability without post-processing has not been achieved.
- **Gradient magnitude**: the continuous adjoint provides correct descent direction (correlation ≥ 0.80) but not the exact gradient magnitude; a discrete adjoint is future work.
- **Mesh convergence**: RMSE is not monotone under identical-schedule refinement; the study characterises optimizer sensitivity, not classical Richardson-extrapolation convergence.
- **Binary performance**: binary-thresholded RMSE (0.359) matches the Christmas-tree baseline (0.329); no manufacturable improvement over existing designs is demonstrated.
- **2D steady flow only**: 3D extension and unsteady flows are implemented but not systematically validated.
- **No experimental validation**: all results are computational; fabrication and testing are necessary next steps.

6.5 Future directions

Immediate priorities are: (i) adaptive mesh refinement targeting the fluid–solid interface; (ii) integration of body-fitted Navier–Stokes post-processing into the main pipeline; (iii) a systematic Pareto front study over objective weights; (iv) three-dimensional validation; and (v) experimental collaboration for a representative subset of designs.

7 Conclusion

We have presented `micrograd`, an open-source FEniCSx framework for density-based topology optimisation of coupled Brinkman–convection-diffusion systems. The key methodological contributions are:

- a complete continuous adjoint derivation for the coupled system, with consistently implemented chain-rule propagation through the Helmholtz filter and Heaviside projection, with gradient direction verified by gradient-descent monotonicity (Section 4.1);

- consistent SUPG stabilisation for both the forward and adjoint transport equations, applied for numerical stability;
- a smooth C^∞ penalty replacing non-differentiable concentration clipping, ensuring adjoint consistency;
- a hybrid OC/MMA optimisation strategy with phased $\alpha_{\max}$ ramping and β-continuation that ensures reliable convergence; and
- full reproducibility via Docker, continuous integration, and Zenodo archival.

As a proof-of-concept, the framework achieves an outlet RMSE of 0.058 for a linear concentration target and reduces hydraulic resistance by 99.7% relative to the classic Christmas-tree baseline. The optimised design achieves a gray-zone fraction of 0.074 = 0.074 (reduced from 0.443 at $\beta = 16$ to 0.074 at $\beta = 128$ via 1400-iteration continuation), reflecting the physical reality that a linear mixing gradient is achieved by distributed porous transport rather than sharp binary channels; body-fitted binary validation is an identified direction for future work.

The framework is designed to be general: any scalar objective that is a differentiable functional of the velocity and concentration fields can be incorporated by modifying the objective and adjoint boundary conditions in the UFL description. This generality, combined with a computational cost of under thirty minutes on a laptop and full reproducibility, makes `micrograd` a practical foundation for computational microfluidic inverse design.

8 Data and code availability

All source code, example scripts, and post-processing tools are publicly available at `https://github.com/nisongmonyimba278-byte/micrograd` under the MIT licence. The repository includes the complete pipeline (forward solver, adjoint solver, Taylor-test module, optimiser, post-processing), a Jupyter notebook for interactive exploration, a test suite, and a continuous-integration workflow (GitHub Actions [16]) that executes a three-stage validation pipeline on every commit inside the official `dolfinx/dolfinx:v0.7.3` Docker container.

All results in this paper are reproducible with a single command:

```
docker run --rm -v $(pwd):/root/micrograd \
  -e OMP_NUM_THREADS=1 -e MPLBACKEND=Agg \
  dolfinx/dolfinx:v0.7.3 bash -c \
  "cd /root/micrograd && bash run_all.sh"
```

The code is available at the GitHub repository above and will be archived on Zenodo upon acceptance (DOI registration pending; [15]).

Supplementary Information

S1: Mesh Convergence and Diffusion-Layer Resolution

The optimisation was repeated on four structured meshes: 20×5, 40×10, 80×20 (primary), and 160×40 (validation), all for the linear gradient target Both 80×20 and 160×40 use identical 600-iter schedules.

Table 4: Optimizer performance across mesh resolutions. The 20×5 mesh is excluded (too coarse; reversed-flow local minima). The 40×10 result shows optimizer sensitivity (plateau after iter 100). The 80×20 primary result uses 600 iterations; the 160×40 validation uses 800 iterations (tuned schedule) (600 iter identical schedule; J monotone, RMSE non-monotone). RMSE measures the global outlet profile error.

Mesh	Elements	$\Delta y\,(\mu\text{m})$	RMSE	Note
20×5	100	100	—	excluded (too coarse; $N_{\text{out}} = 6$)
40×10	400	50	0.414	optimizer not converged
80×20	1600	25	0.058	Primary result
160×40	6400	12.5	0.091	600 iter, identical; J monotone

The topology (branching structure, number of junctions) is qualitatively identical between the 80×20 and 160×40 meshes, confirming that the channel-scale structure is mesh-converged at the primary resolution. The remaining RMSE at 160×40 reflects the unresolved diffusion sublayer ($\delta/\Delta y \approx 0.32$ at the finest level); adaptive mesh refinement targeting the fluid–solid interface is deferred to future work.

For larger meshes, the direct MUMPS solver is replaced by a block-triangular field-split preconditioner: algebraic multigrid (HYPRE) for the velocity block and the Schur complement approximation for the pressure [12]. Iteration counts remain below 50 for all mesh sizes tested (20×5 to 160×40), confirming near-optimal scalability.

S2: Filter Radius Sensitivity

The Helmholtz filter radius $r_f = 2\Delta x$ was varied between $1\Delta x$ and $4\Delta x$ on the 80×20 mesh. The outlet RMSE varied by less than 0.02 across this range, and the qualitative topology was preserved throughout. Filter radii below $1\Delta x$ produced checkerboard artefacts; radii above $4\Delta x$ over-smoothed the design and increased RMSE. The value $r_f = 2\Delta x$ lies in the centre of the stable range and is used throughout the main paper.

S3: SUPG Stabilisation Validation

Three forward solves on the 80×20 mesh were compared for the linear gradient target: (i) no stabilisation, (ii) SUPG with τ from Eq. (28), and (iii) full upwinding ($\tau = h/(2\|\mathbf{u}\|)$). Without stabilisation, the concentration field exhibits oscillations with amplitude up to 0.4 at the inlet interface. SUPG reduces oscillation amplitude below 10^{-3} while maintaining the smooth outlet gradient profile. Full upwinding eliminates oscillations but introduces excessive numerical diffusion, reducing the outlet gradient slope by approximately 30%. SUPG is therefore the appropriate choice for this operating regime.

S4: OC vs. MMA Convergence Comparison

Both the Optimality Criteria (OC) method and the Method of Moving The hybrid OC/MMA schedule was run for 600 iterations on the linear gradient target (80×20 mesh, identical initialisation and continuation schedules). The hybrid OC/MMA strategy (OC for topology formation at $\beta = 1$–2, MMA for sharpening at $\beta = 4$–16) converges to $J = 1.76 \times 10^{-3}$ with outlet RMSE $= 0.058$ in 600 iterations.

Pure OC was tested independently and exhibits a period-2 oscillation at $\beta \geq 4$: the objective alternates between $J \approx 3.6 \times 10^{-3}$ and $J \approx 5.9 \times 10^{-3}$ on every iteration, converging to neither value. Diagnosis revealed that the sensitivity vector has a near-symmetric sign distribution (47% negative, 53% positive), causing the OC bisection to find two complementary designs that it alternates between indefinitely. MMA's asymptote-based move control breaks this cycle by adaptively contracting asymptotes in oscillating directions.

The self-contained Svanberg dual-bisection MMA implementation (`micrograd/optimizer.py`) requires no external packages. The volume constraint is enforced explicitly via dual bisection in the MMA subproblem, achieving $|V - V^*| < 10^{-3}$ at every iteration once the topology is established.

S10: Adjoint Gradient Verification Details

Two verification tests are reported in Section 4.1.

FD Taylor remainder. The finite-difference directional derivative $\hat{g}[\delta\rho] = (J(\rho + \varepsilon\delta\rho) - J(\rho))/\varepsilon$ at $\varepsilon = 10^{-5}$ is used as the reference. The Taylor remainder $R(\varepsilon) = |J(\rho + \varepsilon\delta\rho) - J(\rho) - \varepsilon\hat{g}[\delta\rho]|$ is computed for $\varepsilon \in \{10^{-5}, \ldots, 10^{-2}\}$. Least-squares slopes: 2.12 (20×5), 2.07 (80×20).

Direction test. The mass-scaled adjoint g/m_i is compared to the finite-difference gradient $\hat{g}_i$ component-wise on 20 sampled DOFs. Pearson correlations: 0.80 (20×5), 0.88 (80×20). The magnitude discrepancy reflects the difference between lumped and full mass-matrix Riesz maps; only the direction is used by OC/MMA.

Continuous vs. discrete adjoint. The assembled sensitivity is a continuous adjoint $g_i = \int(\partial J/\partial\bar{\rho})\psi_i \, dx$, not the discrete gradient $\hat{g}_i = \partial J/\partial\bar{\rho}_i$. The relative error $E_g = \|g/m - \hat{g}\|/\|\hat{g}\| = 18.9$ (20×5) reflects this mismatch. A discrete adjoint is deferred to future work.

S5: Thresholding Robustness and Binary Gap Analysis

The optimised design has a grey-zone fraction of 0.074 (measured at $\beta = 64$, 1000-iteration continuation) at $\beta = 16$, reflecting the physical reality that the linear gradient target is achieved by distributed diffusive mixing through intermediate-density Brinkman elements. Hard-thresholding at $\bar{\rho} = 0.5$ and re-solving the forward problem gives the results in Table 5.

Table 5: Binary validation results on the 80×20 mesh. Both binary configurations give identical RMSE, confirming the gap is caused by loss of gray mixing pathways, not by the Brinkman penalisation in fluid regions.

Design	**RMSE**	**ΔRMSE vs grey**	**Method**
Grey Brinkman	0.058	—	Smooth $\alpha(\bar{\rho})$
Binary Brinkman	0.359	+356%	$\bar{\rho} \in \{0, 1\}$, smooth α
Binary Stokes	0.359	+356%	$\bar{\rho} \in \{0, 1\}$, $\alpha = 0$ in fluid

The + + 356% RMSE change is mechanistically expected: the linear gradient target $c^\star(y) = y/L_y$ is optimally achieved by diffusive transport through intermediate-density elements, which have no binary physical analogue. The identical results for binary Brinkman and binary Stokes confirm that the gap is not caused by the Brinkman penalisation in fluid regions (setting $\alpha = 0$ makes no difference), but by the loss of the gray mixing pathway. The analytical bound $t_{\text{diff}}/t_{\text{flow}} \approx 3000$ (Section 2.3) confirms that solid leakage through $D_{\min}$ is negligible; the binary gap is a property of the topology, not a numerical artefact. Body-fitted Navier–Stokes validation on a purpose-designed binary multi-channel manifold is deferred to future work.

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